2412ET040

Roll No.

Course: B.Tech Computer Engineering /
B.Tech COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)
Subject: Database Management Systems
Subject Code: ETCS-205
Semester: III

Time: 03 Hours **Max Marks: 70**

Instructions to the Students:

1. This Question paper consists of two Sections. All sections are compulsory.
2. Section A comprises 10 questions of short answer type. All questions are compulsory. Each question carries 2 marks.
3. Section B comprises 8 long answer type questions out of which students must attempt any 5. Each question carries 10 marks.
4. Do not write anything on the question paper.

Q.No.	SECTION –A (SHORT ANSWER TYPE QUESTIONS)	Marks
1. a.	State the various abstraction level of database.	(2)
b.	What is dirty read problem?	(2)
c.	Why we use Armstrong Axioms? List any one rule with the help of example.	(2)
d.	Write down 2 features of Relational calculus.	(2)
e.	Write down two points of differences between MySQL and IBM Db2	(2)
f.	Define relationship degree in DBMS.	(2)
g.	What is normalization?	(2)
h.	Write an SQL for displaying current date.	(2)
i.	List and briefly explain two binary relational operations.	(2)
j.	List four characteristics of deadlock ?	(2)

SECTION –B (LONG ANSWER TYPE QUESTIONS)

2. Describe the various database management system architectures with suitable examples. (10)

3. Explain following terms by giving suitable examples- (10)
   i. Data Definition language
   ii. Data Manipulation language
4. Differentiate between generalization and specialization in details by taking suitable example. (10)
5. Sketch an ER model of University/Educational Institute management system. The model should include - (a) courses (b) Program Offered (c) students (d) Faculty, Further, the enrolment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modelled. Document all assumptions that you make about the mapping constraints. (10)
6. Differentiate between following: - (10)
   Internal and external Hashing
   Primary and Secondary Indexes
7. What is deadlock in DBMS? Explain various deadlock prevention mechanism. (10)
8. Write short note on Normalization. Specify the rules of following normal forms - (10)
   - 1NF
   - 2NF
9. Explain various types two phase locking technique for concurrency control in details using suitable example. (10)

===END OF PAPER===